

Response Letter with Annotations

Waller, I., & Anderson, A. (2021). Quantifying social organization and political polarization in online platforms. *Nature*, 600(7888), 264–268. <https://doi.org/10.1038/s41586-021-04167-x>

Annotations

1. Purposes

- a. Motivation: Twenty years into the widespread adoption of online social media platforms, it remains unclear how online communities are socially organized.
- b. Specific Rationale (several bullet points)
 - Prior studies focused on:
 1. Biases that result from using self-reported data, expert labels and survey-based methods by quantifying the social makeup of communities in a purely behavioural fashion.
 2. Previous analyses have studied platforms such as Facebook, Twitter and Amazon, on which users are guided by algorithmic curation and personalized recommendations.
 3. Existing empirical research is largely limited to static analyses of incomplete and non-representative snapshots of activity on a platform
 - Lack of:
 1. Empirical data quantifying the social makeup of online communities and, in turn, the social organization of online platforms, poses a unique challenge.
 2. Although there is evidence that online platforms exist in states of partisan fragmentation, important questions about the dynamics and mechanisms of polarization processes remain unanswered.
 - Need to:
 1. Develop and validate a methodology using neural community embeddings, which represent similarities in community membership as relationships between vectors in a high-dimensional space, to quantify the positioning of online communities along social dimensions.
 2. Apply it here to Reddit, which has maintained a minimalist approach to personalized algorithmic recommendation throughout its history.
 3. Conduct dynamic analyses of complete platform activity to measure both platform- and individual-level polarization, and compare these for the left and right wings, over the entire history of Reddit.
- c. Research Questions and/or Hypotheses
 - To what extent does platform-level political polarization change over time?

- Do individual users become more polarized in their political activity over time, and if so, do these changes drive platform-level polarization?
- Are the dynamics of polarization ideologically symmetric?
- **Identify IV, DV, mediator, moderator**
 1. **IV:** Time and external events (specifically focused on the 2016 US presidential election as a major external catalyst).
 2. **IV2:** User cohorts (grouped based on date of their first political comment).
 3. **IV3:** Social dimensions (seed pairs; *r/democrats* and *r/Conservative*) as independent anchor points for the dimensions of age, gender, and partisanship
 4. **DV:** Macroscale social organizations (high-level stratification measured by community cluster along social lines)
 5. **DV2:** Community scores (assigned every community a specific z-score on dimensions of age, gender, and partisanship).
 6. **DV3:** Platform-level political polarization (mean absolute partisan z-score of political comments)
 7. **DV4:** Individual-level polarization (shift in individual user's political activity toward partisan extremes over time)
 8. **Mediators:** New user arrival (population turnover) and individual users moving toward extremes
 9. **Moderators:** Ideological orientation (left vs right) and community topic/cluster (type of community; programming vs. personal matters)
- Identify specific relationships among variables
 1. Increase in polarization → driven more by new users than individual change
 2. Right-wing activity → stronger contributor to polarization
 3. External events (e.g., elections) → aligned with polarization spikes

2. Method

a. Research Design

- Computational study
- Neural community embeddings

b. Sample

- The complete set of comments from Reddit (2005–2018)
- 5.1 billion comments
- 10,000+ Reddit communities
- Covers ~95% of platform activity
- 95.4% of all comments and 93.2% of all users
- 34.7 million Reddit commenters: complete public commenting history, the communities (subreddits), and the timestamps associated with each comment.
- Over our entire study period, 52.9% of users commented in more than one subreddit, and the mean number of one user = 9.6.

c. Data collection

- The data was obtained from pushshift.io Reddit archive, a third-party repository that collects and stores Reddit activity
- Authors utilized three pieces of information: username, name of the community, and timestamp.
- Deleted comments were excluded.

d. Measures

- **Social dimensions (age, gender, and partisanship)**
 1. Neural community embeddings to represent 10, 006 communities as vectors in a 150-dimensional space
 2. Social dimensions like partisan were identified by using a seed pair (e.g., *r/democrats* and *r/Conservative* and *r/AskMen* and *r/AskWomen* for gender).
 3. Seed pair was augmented with nine additional similar pairs to ensure robustness ($k = 10$). Vector differences were averages to create a final vector
 4. Every community was assigned a score by projecting its vector onto the social dimension vector. Reflects average similarity to the communities at one pole (right wing) minus its similarity to the other pole (left-wing).
- **Macroscale platform organization:**
 1. The authors applied hierarchical clustering to the community embedding to group subreddits by behavioral similarity.
 2. Distribution of the clusters were analyzed along the measure social dimensions to see if the platform was stratified by age, gender, or politics.
- **Political polarization:**
 1. Measured as both state and process using the mean absolute partisan z-score of political comments.
 2. Scores were tracked from 2012 to 2018 to identify shifts (sharp polarization) during the 2016 presidential elections
 3. Mechanisms were distinguished by grouping users into cohorts based on the date of their first political comment.
 4. Measured if individual users moved toward extremes or if new users arrived to drive platform-level changes.
- **Partisan-ness:**
 1. Calculated by taking the sum of seed pair vectors instead of the difference.
- **External validations:**
 1. Community scores were validated against real-world data (comparing gender scores for occupation-related subreddits against US Census Bureau gender makeup statistic)
 2. Comparing partisan scores for city-based subreddits against 2016 US presidential election results.

e. Analysis

- Community embeddings: are learned solely from interaction data—high similarity between a pair of communities requires not a similarity in language but a similarity in the users who comment in them.
 1. Open source software word2vecf: word2vec algorithm to interaction data by treating communities as ‘words’ and users as ‘contexts’—every instance of a user commenting in a community becomes a word–context pair.
 2. Skip-gram with negative sampling (SGNS) method.
 3. Grid search of hyperparameters (sample, the number of negative examples; alpha; and size, a Boolean parameter which indicates whether the training data should be randomly shuffled prior to training).
 4. Model optimization using community analogies
- Agglomerative hierarchical clustering techniques (Euclidean distance) to partitions the communities into 30 topical clusters
- t-SNE projection: t-distributed stochastic neighbor embedding to project the 150-dimensional space into a two-dimensional visualization
- Longitudinal cohort analysis
- Statistical comparisons over time
- Pearson correlation coefficients (compare algorithmically generated community scores against external real-world data)
- Analyses were performed with Python v3.7, pandas v1.3.3 and Spark v3.0.
-

3. Results and Conclusions

- a. Answers to RQs and Hypotheses
 - Reddit structure is stratified along social lines (age, gender, and partisanship).
 - Platform-level polarization increased sharply around 2016 due to the elections
 - Individual users rarely become polarized over time
 - Polarization is driven by new users entering the platform
 - Polarization is asymmetric, driven largely by right-wing activity
- b. Include Key Figures/Tables

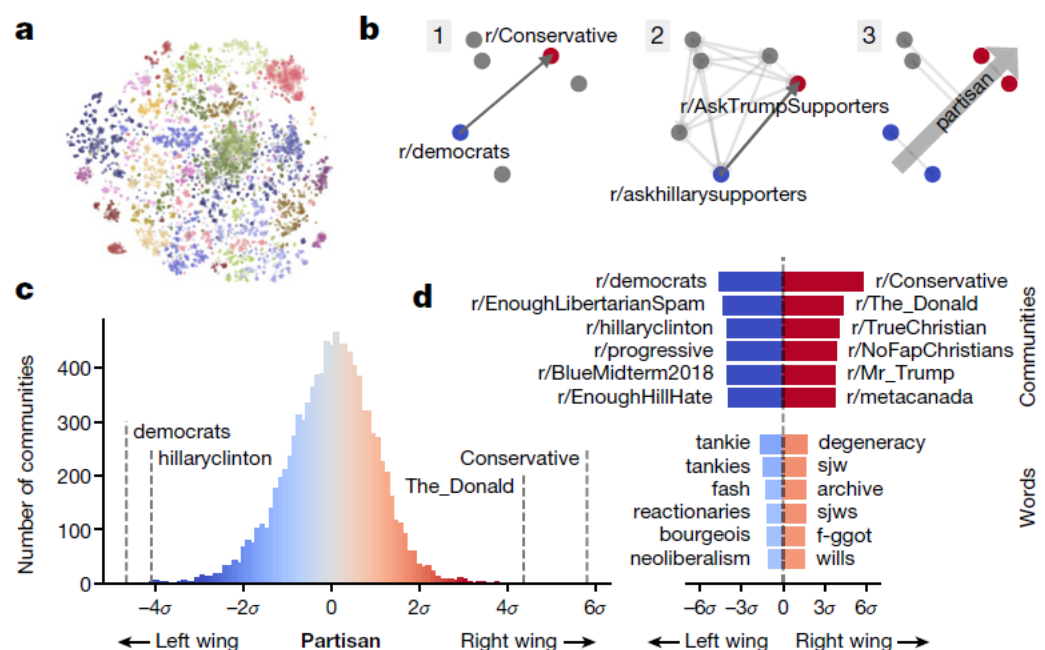
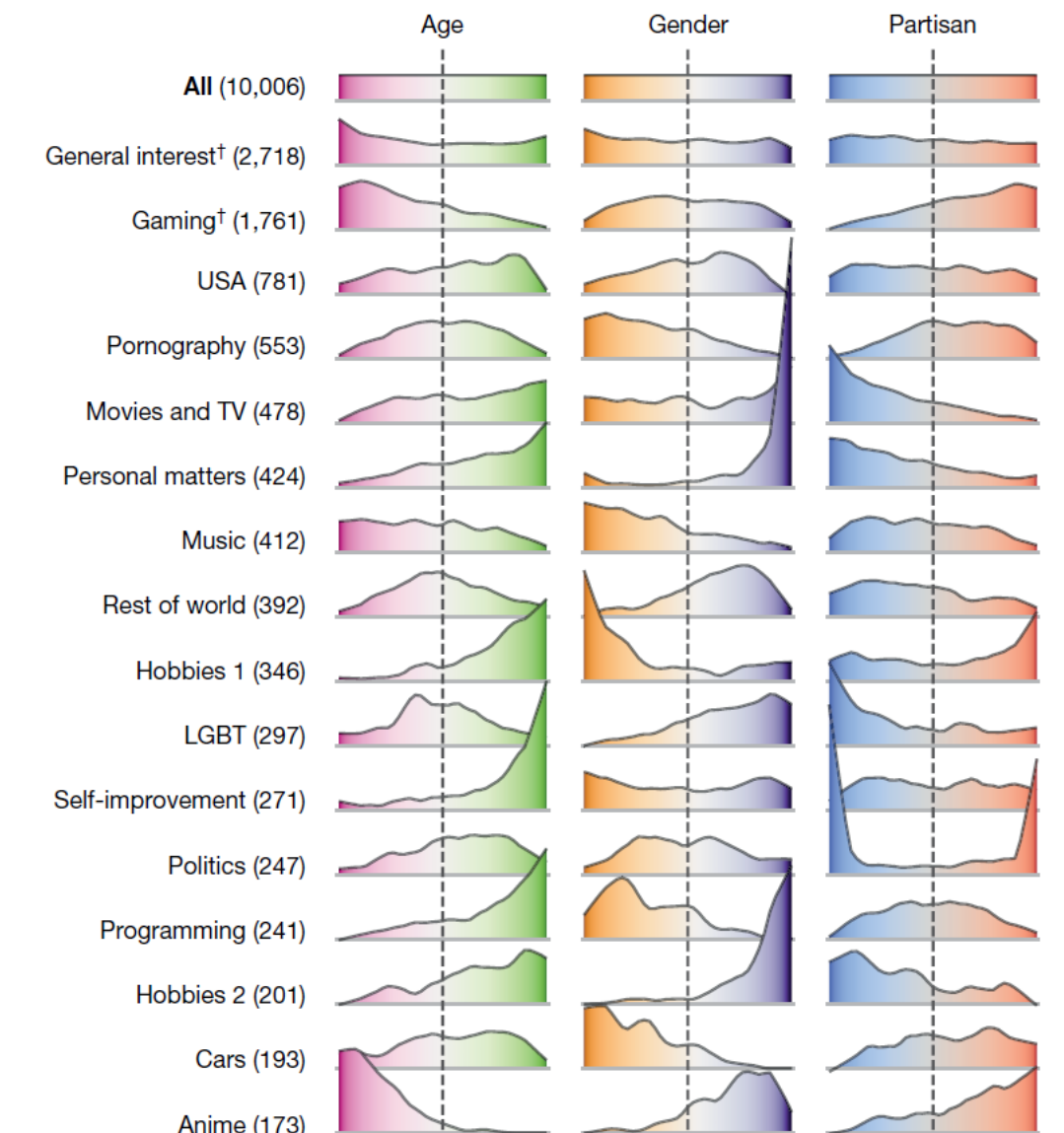


Fig. 1 | Quantifying social dimensions on Reddit. **a**, A two-dimensional t -distributed stochastic neighbour embedding (t -SNE) projection of the 10,006 subreddits in our Reddit community embedding, with points coloured by clusters found by hierarchical clustering. **b**, An illustration of our methodology to generate social dimensions. **c**, The distribution of partisan scores for the 10,006 most popular Reddit communities. The x-axis shows the number of standard deviations from the mean partisan score (z -score). Communities vary from far-left wing to far-right wing and are coloured by z -score. **d**, Top, communities most associated with the left-wing and right-wing ends of the dimension (for community descriptions, see Supplementary Table 1). Bottom, words most associated with the left-wing and right-wing ends of the dimension, considering only word usage in political communities in 2017 as quantified by the partisan-ness dimension (Extended Data Fig. 6).



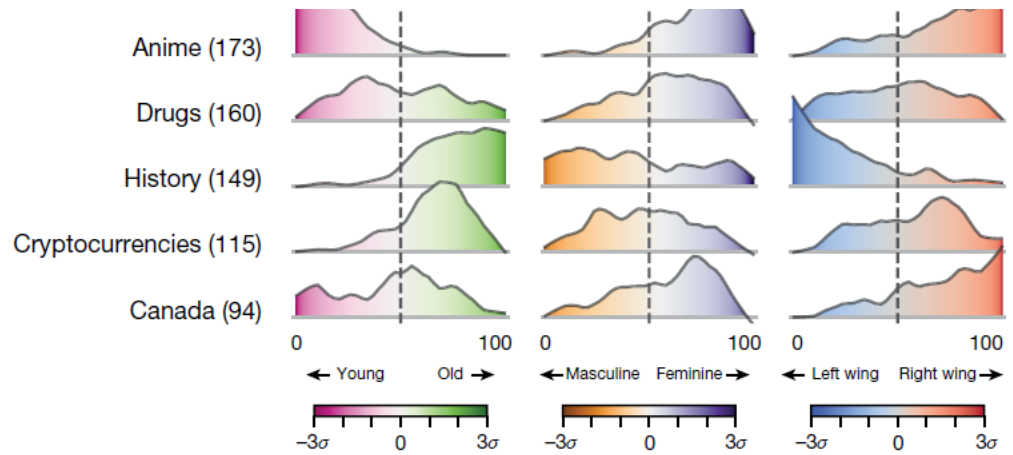


Fig. 2 | Macroscale social organization of Reddit communities.

Distributions of communities along the age, gender and partisan dimensions, grouped into behavioural clusters found by hierarchical clustering. The x-axis represents community scores transformed into percentiles (for example, a community with age score greater than 76% of other communities would be positioned at percentile 76), and colour corresponds to z-score. As a result, the distribution for all communities (top row) is simply the uniform distribution $U(0, 100)$, while the distributions for individual clusters illustrate which percentiles are over- or under-represented within the cluster. Raw score (non-percentile) distributions are shown in Extended Data Fig. 4. The dashed line indicates the 50th percentile. Rows annotated with † comprise two or more clusters (Methods, 'Creating the community embedding').

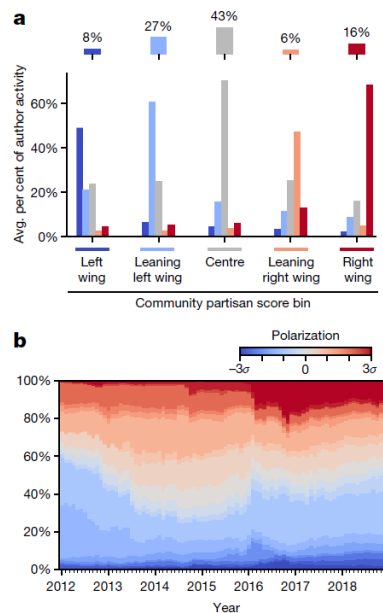


Fig. 3 | Distribution of political activity on Reddit. a, Average (avg.) distributions of political activity contributed by users of different partisan community bins. The top distribution shows the average distribution for all users—that is, independent of partisan activity—while each of the five bottom distributions shows the average distribution of political activity contributed by users who commented in the corresponding partisan category. **b,** The distribution of political activity on Reddit over time by partisan score. Each bar represents one month of comment activity in political communities on Reddit and is coloured according to the distribution of partisan scores of comments posted during the month (where the partisan score of a comment is the partisan score of the community in which it was posted.).

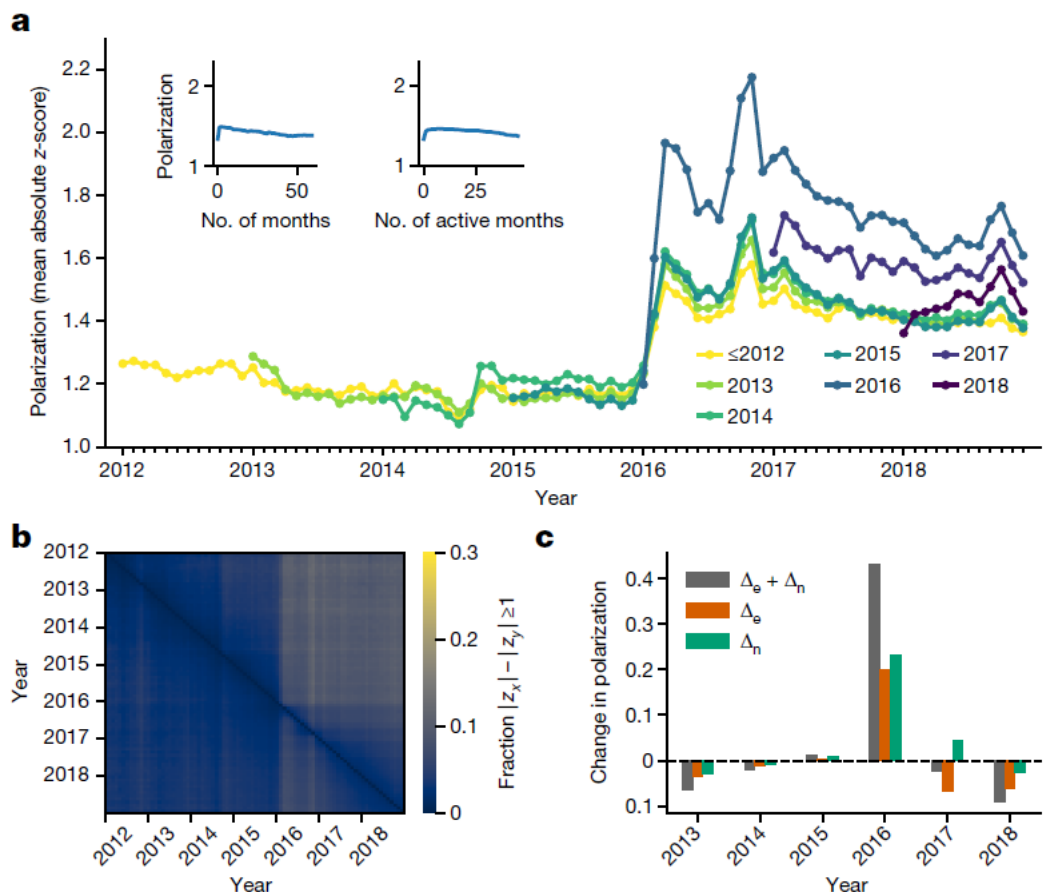


Fig. 4 | Political polarization of new and existing users. **a**, The average polarization of political activity on Reddit, broken down into seven user cohorts by year of the author's first political activity. Polarization is measured by the absolute z-score of the community—that is, the absolute number of s.d. from the mean partisan score. Inset, the relationship between polarization and number of total (left) and active (right) months since a user's first political activity. **b**, Observed within-user polarization. Each (x, y) cell represents the proportion of users whose average polarization (average absolute z-score) increased by one s.d. between months t_x and t_y (minor ticks on the x -axis indicate month), for all users active during both months. A user is considered active if they make at least 10 comments in a month. Results are robust for other choices of these two thresholds (Extended Data Fig. 7f, g). **c**, Annual change in polarization, decomposed into the change attributable to new (Δ_n) and existing (Δ_e) users. The grey bar represents the actual observed year-over-year change in polarization; Δ_e represents the change that would be observed had new users not changed at all (that is, they were only as polarized as the overall polarization in the previous year); Δ_n represents the change that would be observed had existing users not changed at all (that is, they remained only as polarized as in the previous year).

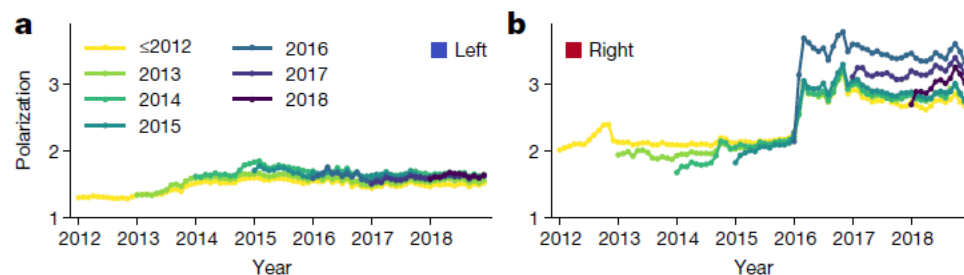


Fig. 5 | Ideological asymmetry in online polarization. **a, b**, Average polarization of activity in the left (**a**) and right (**b**) wings, decomposed into seven cohorts by year of first political activity.

c. Authors' Conclusions

- Online polarization is not primarily due to individuals becoming more extreme
- Instead, it is driven by. population changes (new users) and external events
- Platform structure reflects **behavioral aggregation**, not just algorithmic effects

4. Discussion

a. Implications

- Challenges the idea that social media “radicalizes” individuals
- Suggests polarization is driven by who joins the platform and external political events
- Provides a scalable method to study online social structure

b. Limitations

- Assumes community characteristics are stable over time
- Cannot fully capture, Multi-account (“throwaway”) behavior
- Limited to Reddit (platform-specific)
- Measures social constructs indirectly (embeddings $\neq$ true identity)

c. Future Directions

- Neural-embedding methodology to be used on other online platforms
- Model could be expanded to include more complex social constructs, such as using multiple dimensions to represent gender and other social notions of identity such as class and affluence

--- Page Break Here ---

Response Letter (~600 words)

Assessment of the study

Waller and Anderson (201) present an innovative computational approach to understanding social organization and political polarization in online platforms. Using a massive dataset of Reddit comments and a novel neural embedding methodology, the authors challenge conventional assumptions about how polarization develops online. They find that platform-level polarization is driven by new users, particularly around major political events such as the U.S. presidential elections. This is oppositional to studies that have found that individuals become increasingly polarized over time. This broadly speaks to what we have been discussing about social media, online echo chambers, and the dynamics of information environments.

Strengths

One notable strength of the paper is the complex computational approach used. By leveraging 5.1 billion comments across over 10,000 communities, the authors are able to analyze social organization at an unprecedented scale. Particularly fascinating is the use of neural embeddings. This allowed the authors to infer social dimensions, such as political orientation, solely on behavioral data rather than self-reports or surveys. It avoids common biases associated with self-reported identity and enables large-scale data analysis. The longitudinal nature of the study also allows the authors to distinguish between individual-level and platform-level changes over time.

Weaknesses

Although the computational method used comes with its strengths, we can identify several limitations. The interpretation of embedding-based measure raises conceptual concerns.

While the authors treat positions in the embedding space as indicators of social constructs, like political orientation, these are indirect proxies. It is not entirely clear how accurately these representations capture real-world identities or beliefs. The study also assumes that community relationships remain relatively stable over time, which may overlook important shifts in community composition.

Although Reddit is less algorithmically curated than other platforms, it is not entirely free from platform effects. The study cannot fully account for how platform design or recommendation systems might influence user behavior. Although neural embeddings are appealing to use, they are complex and not easily interpretable, which makes it difficult to fully understand how the model derives its results.

Other notes

Overall, this study makes a significant contribution to both communication research and the broader understanding of social organization and polarization. Studies have highlighted how social media contribute to echo chambers and push individuals to have extreme views. This study complicates the narrative by demonstrating that large-scale polarization may emerge from population turnover rather than individual radicalization. Overall, Waller and Anderson provide an example of how computational methods can uncover large-scale social patterns.

The more I read articles that have employed computational methods, the more I feel that I have a long way to go. Scholars are using complex methods to understand human behavior and that helps us understand complex situations and relationships. Combined with other complementary approaches, we can better capture underlying mechanisms of human behavior.